

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME:CSA1610 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

<i>Criteria</i>	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _G.Hima Bindu(192511377)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Hima Bindu

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

1. Introduction

K-Means and **Hierarchical Clustering** are unsupervised data mining techniques used to group similar customers based on their characteristics.

For the given dataset, customers are analyzed using two attributes:

- **Customer Income (₹)**
- **Spending Score**

The objective is to identify customer segments and compare the performance and scalability of the two clustering techniques.

2. Dataset

Customer	Income (₹)	Spending Score
C1	30,000	60
C2	60,000	40
C3	50,000	70
C4	28,000	65
C5	65,000	35

Before applying clustering, the attributes should preferably be **normalized**, because income values are much larger numerically than spending scores. This prevents income from dominating distance calculations.

3. K-Means Clustering

K-Means is a **partition-based clustering algorithm**. It divides the dataset into a predefined number of clusters.

The basic steps are:

1. Select the number of clusters, K .
2. Initialize K centroids.
3. Calculate the distance between each customer and the centroids.
4. Assign each customer to its nearest centroid.
5. Recalculate the centroids.
6. Repeat the process until the clusters become stable.

For example, if $K = 2$ is selected, the customers can be broadly divided into:

- **Lower-income group:** C1 and C4
- **Higher-income group:** C2, C3 and C5

However, because spending score is also considered, the exact cluster membership depends on the distance calculation after normalization.

Advantages of K-Means

- Simple and easy to implement.
- Computationally fast.
- Works well with large datasets.
- Requires relatively low memory.
- Suitable for customer segmentation.

Limitations

- The value of K must be selected in advance.
- Sensitive to the initial centroid selection.
- Sensitive to outliers.
- Works best when clusters are relatively compact and well separated.

4. Hierarchical Clustering

Hierarchical clustering creates a **hierarchy of clusters** rather than directly producing a fixed number of clusters.

In **Agglomerative Hierarchical Clustering (HAC)**, each customer initially forms an individual cluster. The algorithm repeatedly merges the most similar clusters.

For this dataset, customers with similar income and spending behavior are merged first. The process continues until all customers belong to one cluster.

The result is represented using a **dendrogram**.

A horizontal cut through the dendrogram can be used to select the required number of customer segments.

Advantages of Hierarchical Clustering

- Does not require the number of clusters to be fixed initially.
- Produces a dendrogram showing relationships between customers.
- Useful for exploring the natural structure of data.
- Can provide more detailed hierarchical information.

Limitations

- Computationally more expensive than K-Means for large datasets.
- Requires more memory.
- Once two clusters are merged, the decision generally cannot be reversed.
- Less suitable for very large customer databases.

5. Comparison of the Two Methods

Feature	K-Means	Hierarchical Clustering
Type	Partition-based	Hierarchical
Cluster number	Must specify K	Can determine using dendrogram
Output	Fixed clusters	Dendrogram + clusters
Speed	Generally faster	Generally slower
Scalability	High	Lower
Memory requirement	Relatively low	Relatively high
Large datasets	Suitable	Less suitable
Outlier sensitivity	High	Depends on linkage/distance
Initialization	Centroids required	No centroid initialization
Visualization	Less informative	Dendrogram provides hierarchy
Flexibility	Less flexible	More exploratory
Best use	Large-scale segmentation	Exploring relationships and hierarchy

6. Performance Comparison

For the given dataset containing only **5 customers**, both K-Means and hierarchical clustering can be executed easily.

K-Means generally performs faster because it repeatedly assigns points to centroids and updates them. Hierarchical clustering requires calculating and updating distances between clusters during the merging process.

Therefore, for this small dataset, the performance difference may not be significant.

However, as the number of customers increases, hierarchical clustering becomes more computationally expensive, whereas K-Means generally remains more practical.

7. Scalability Comparison

Scalability refers to how effectively an algorithm handles increasing amounts of data.

K-Means

K-Means is highly scalable and can be applied to large customer datasets containing thousands or millions of records. Its computational requirements are generally lower than those of hierarchical clustering.

Hierarchical Clustering

Hierarchical clustering requires distance calculations and maintains a hierarchy of clusters. As the dataset becomes larger, the computational and memory requirements increase significantly.

Therefore:

K-Means → Better scalability

Hierarchical Clustering → Better for smaller datasets and exploratory analysis

8. Customer Segmentation Interpretation

The dataset suggests different customer behavior patterns.

- **C1 and C4:** Lower-income customers with relatively high spending scores.
- **C2 and C5:** Higher-income customers with relatively low spending scores.
- **C3:** Higher-income customer with a high spending score.

This shows that **income alone does not determine spending behavior**. Clustering considers both income and spending score to identify meaningful customer segments.

A company can use these segments for:

- Targeted marketing
- Personalized offers
- Customer profiling
- Product recommendations
- Loyalty programs
- Advertising strategies

For example, customers with high spending scores can be targeted with premium offers, while high-income but low-spending customers may require different promotional strategies to increase engagement.

9. Which Method is More Suitable?

For a **small dataset**, hierarchical clustering can be useful because the dendrogram provides a visual representation of customer relationships and helps identify the natural number of clusters.

For a **large e-commerce or retail dataset**, K-Means is generally more suitable because of its better computational efficiency and scalability.

Therefore, the choice depends on the objective:

- **Small dataset + relationship exploration → Hierarchical Clustering**
- **Large dataset + fast customer segmentation → K-Means**

10. Conclusion

Both K-Means and hierarchical clustering can be used to segment the given customers based on **income and spending score**.

K-Means is a **fast, simple, and scalable partition-based method**, but it requires the number of clusters to be specified beforehand. Hierarchical clustering builds a **hierarchy of customer groups** and provides a dendrogram, making it useful for exploring relationships and deciding the number of clusters.

For the given small dataset, both methods are practical. However, for large-scale customer segmentation, **K-Means is generally preferred because of its better performance and scalability**, while hierarchical clustering is more suitable for detailed exploratory analysis.

Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark)

1. Introduction

K-Means and **Expectation-Maximization (EM)** are clustering techniques used in data mining to group similar data objects.

In this dataset, each document is represented using three features:

- Word1
- Word2
- Word3

The document vectors are:

Document	Word1	Word2	Word3
D1	0.20	0.50	0.30
D2	0.10	0.60	0.30
D3	0.80	0.10	0.10
D4	0.75	0.15	0.10
D5	0.20	0.40	0.40

The objective is to compare how K-Means and EM handle documents that may have similarities with more than one cluster.

2. Interpretation of the Dataset

The documents show two major patterns:

- **D1 and D2** have high Word2 values and relatively low Word1 values.
- **D3 and D4** have very high Word1 values and low Word2 and Word3 values.
- **D5** has a combination of Word1, Word2 and Word3 values, making it potentially closer to the first group while also having characteristics different from D1 and D2.

Therefore, the dataset contains relatively clear groups as well as a possible **overlapping/borderline document D5**.

A possible conceptual grouping is:

Cluster 1: D1, D2, D5

Cluster 2: D3, D4

The important question is how each algorithm represents the membership of D5.

3. K-Means Clustering

K-Means is a **hard clustering algorithm**. It assigns each document to exactly one cluster based on its distance from the cluster centroids.

The basic procedure is:

1. Select the number of clusters, K.
2. Initialize K centroids.
3. Calculate the distance between each document and each centroid.
4. Assign each document to its nearest centroid.
5. Recalculate the centroids.
6. Repeat until the clusters become stable.

For example, if **K = 2**, K-Means may produce:

Document	K-Means Assignment
D1	Cluster 1
D2	Cluster 1
D3	Cluster 2

D4	Cluster 2
D5	Cluster 1

Thus, D5 is forced to belong completely to one cluster.

Limitation for Overlapping Data

K-Means does not express uncertainty or partial membership.

If D5 is almost equally similar to both clusters, K-Means still has to make a single decision:

D5 → Cluster 1

It cannot represent:

D5 → 60% Cluster 1 and 40% Cluster 2

Therefore, K-Means is less suitable when document clusters overlap.

4. Expectation-Maximization (EM)

The **Expectation-Maximization algorithm** is a probabilistic clustering approach. Instead of assigning a document completely to one cluster, it can calculate the **probability of membership** in each cluster.

EM works iteratively through two main steps:

E-Step – Expectation

The algorithm calculates the probability that each document belongs to each cluster.

For example, D5 could theoretically receive membership probabilities such as:

Document	Cluster 1 Probability	Cluster 2 Probability
D5	0.65	0.35

This means D5 has a stronger association with Cluster 1 but also has a significant relationship with Cluster 2.

M-Step – Maximization

The algorithm updates the parameters of the clusters using the calculated membership probabilities.

The E-step and M-step are repeated until the model converges.

5. Handling Overlapping Clusters

The major difference between K-Means and EM is how they treat cluster membership.

K-Means

K-Means provides **hard membership**:

D5 → Cluster 1

The document belongs to only one cluster.

EM

EM provides **probabilistic membership**:

D5 → 65% Cluster 1, 35% Cluster 2

Therefore, EM can represent uncertainty and overlapping relationships more effectively.

6. Comparison of K-Means and EM

Feature	K-Means	EM
Clustering type	Hard clustering	Probabilistic/soft clustering
Membership	One cluster	Probability of belonging to each cluster
Overlapping clusters	Handles poorly	Handles effectively
Output	Cluster label	Membership probabilities + cluster parameters
Main concept	Distance from centroid	Probability distribution
Cluster shape	Generally suited to compact clusters	Can model more flexible distributions
Complexity	Relatively simple	More computationally complex
Speed	Generally faster	Generally slower
Uncertainty	Not represented	Explicitly represented
Suitable for ambiguous documents	Less suitable	More suitable

7. Application to the Given Dataset

D1 and D2 have similar word distributions:

- D1 = (0.2, 0.5, 0.3)
- D2 = (0.1, 0.6, 0.3)

Therefore, both algorithms are likely to place them in the same cluster.

Similarly:

- D3 = (0.8, 0.1, 0.1)
- D4 = (0.75, 0.15, 0.1)

are highly similar and are likely to form another cluster.

D5 = (0.2, 0.4, 0.4) is more interesting because its word distribution is different from D3 and D4 but shares characteristics with the D1/D2 group.

With K-Means, D5 would be assigned to whichever centroid is closer.

With EM, D5 can have a **probability of belonging to both clusters**, making the result more informative when document categories overlap.

8. Why EM is Better for Overlapping Documents

In real-world document mining, a document may contain words related to multiple topics.

For example, a document could discuss both:

- Artificial Intelligence
- Healthcare

It would be unrealistic to force such a document completely into only one topic.

EM can represent this using probabilities, such as:

AI cluster = 0.60

Healthcare cluster = 0.40

This provides a more realistic representation of document characteristics.

9. Advantages and Disadvantages

K-Means Advantages

- Simple to understand and implement.
- Computationally efficient.
- Works well with large datasets.
- Suitable when clusters are clearly separated.

K-Means Disadvantages

- Requires K to be specified.
- Produces hard assignments.
- Does not represent uncertainty.
- Less effective for overlapping clusters.
- Sensitive to initial centroid selection.

EM Advantages

- Handles overlapping clusters effectively.
- Provides probabilistic membership.
- Represents uncertainty in cluster assignment.
- More flexible for complex data distributions.

- Useful when an object can belong to multiple groups.

EM Disadvantages

- More computationally expensive than K-Means.
- May require more iterations to converge.
- Sensitive to initialization.
- Assumes an underlying probability distribution/model.

10. Conclusion

For the given document dataset, **K-Means and EM can both identify groups such as {D1, D2, D5} and {D3, D4}**, but they handle overlapping membership differently.

K-Means uses hard clustering, so every document is assigned completely to one cluster. This makes it fast and simple but less suitable for ambiguous documents.

EM uses probabilistic clustering, allowing a document to have different membership probabilities across clusters. Therefore, EM is **more suitable for overlapping document clusters** because it captures uncertainty and provides a more detailed representation of document-topic relationships.

Hence, for this dataset where overlapping membership is important, **EM is preferred over K-Means**, while K-Means is preferable when the clusters are clearly separated and computational efficiency is the main requirement.

Q3. Dataset: Geographic Data

Point Latitude Longitude

P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q3. Comparison of Euclidean and Manhattan Distance Metrics

1. Introduction

Distance metrics are important in **Data Mining and clustering** because they determine how similarity or dissimilarity between data points is measured.

For the given geographic dataset, each point is represented using:

- **Latitude**
- **Longitude**

The two distance metrics considered are:

1. **Euclidean Distance**
2. **Manhattan Distance**

The choice of distance metric can affect which geographic points are considered similar and, consequently, can affect the final clustering results.

2. Given Dataset

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

3. Euclidean Distance

Euclidean distance represents the **straight-line distance** between two points.

For two points:

$$P_i = (x_i, y_i)$$

and

$$P_j = (x_j, y_j)$$

the Euclidean distance is calculated using:

$$d_E(P_i, P_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

Example: Distance between P1 and P2

Given:

$$P_1 = (13.08, 80.27)$$

$$P_2 = (13.10, 80.25)$$

Substituting the values:

$$\begin{aligned} d_E(P_1, P_2) &= \sqrt{(13.08 - 13.10)^2 + (80.27 - 80.25)^2} \\ &= \sqrt{(-0.02)^2 + (0.02)^2} \\ &= \sqrt{0.0004 + 0.0004} \\ &= \sqrt{0.0008} \end{aligned}$$

$d_E(P_1, P_2) \approx 0.0283$

Therefore, the **Euclidean distance between P1 and P2** is approximately 0.0283 degrees.

4. Manhattan Distance

Manhattan distance measures the distance by adding the **absolute differences** between the coordinates.

For two points:

$$P_i = (x_i, y_i)$$

and

$$P_j = (x_j, y_j)$$

the Manhattan distance is:

$$d_M(P_i, P_j) = |x_i - x_j| + |y_i - y_j|$$

Example: Distance between P1 and P2

Given:

$$P_1 = (13.08, 80.27)$$

$$P_2 = (13.10, 80.25)$$

Substituting the values:

$$\begin{aligned} d_M(P_1, P_2) &= |13.08 - 13.10| + |80.27 - 80.25| \\ &= |-0.02| + |0.02| \\ &= 0.02 + 0.02 \end{aligned}$$

$$d_M(P_1, P_2) = 0.04$$

Therefore, the **Manhattan distance between P1 and P2 is 0.04 degrees.**

5. Distance Comparison

Some pairwise distances illustrate the difference between the metrics:

Pair	Euclidean Distance	Manhattan Distance
P1–P2	0.0283	0.0400
P1–P4	0.0361	0.0400
P3–P5	0.0361	0.0400
P1–P5	0.0943	0.1500

The values are different because Euclidean distance measures the direct geometric distance, while Manhattan distance adds the absolute movement along each coordinate

6. Differences Between Euclidean and Manhattan Distance

Feature	Euclidean Distance	Manhattan Distance
Formula	Square root of squared differences	Sum of absolute differences
Movement	Straight-line	Axis-by-axis
Geometry	Circular/spherical neighborhoods in 2D	Diamond-shaped neighborhoods in 2D
Sensitivity to large differences	More sensitive because differences are squared	Less sensitive
Computational complexity	Slightly higher	Simpler
Typical use	Continuous numerical data	Grid-based or high-dimensional data
Effect of outliers	More sensitive	Relatively less sensitive

7. Impact on Clustering

The distance metric directly affects clustering because clustering algorithms use distance to decide which points are similar.

Using Euclidean Distance

Euclidean distance tends to group points based on their **straight-line proximity**.

For this dataset, geographically close points such as **P1 and P2** have a small Euclidean distance and are likely to be placed in the same cluster.

Similarly, **P3 and P5** are relatively close and may form another group.

Using Manhattan Distance

Manhattan distance measures the total coordinate-wise difference. Therefore, the relative distances between points can change compared with Euclidean distance.

This can lead to different cluster boundaries, particularly when points are arranged diagonally or when the data contains outliers.

8. Example of Clustering Impact

Suppose a clustering algorithm such as **K-Means** is applied.

With Euclidean distance, the algorithm may identify groups based on direct geographic proximity:

- **Cluster 1:** P1, P2, P4
- **Cluster 2:** P3, P5

The exact grouping depends on the selected value of K and centroid initialization.

If Manhattan distance is used instead, the distance relationships can change, potentially changing cluster assignments, especially for points that are close to the boundary between clusters.

Thus:

Distance metric → Distance values → Similarity relationships → Cluster assignment

9. Advantages of Each Metric

Euclidean Distance

- Represents direct geometric distance.

- Suitable for continuous numerical data.
- Commonly used with K-Means.
- Easy to interpret.
- Works well when clusters are compact and roughly spherical.

Manhattan Distance

- Simple to calculate.
- Less affected by individual large coordinate differences.
- Useful when movement is constrained along separate dimensions.
- Can be useful for grid-like geographic environments such as city-block navigation.
- Can provide different and sometimes more robust clustering behavior.

10. Important Geographic Consideration

Latitude and longitude are **geographic coordinates**, not ordinary Cartesian coordinates.

Therefore, using Euclidean or Manhattan distance directly on latitude and longitude is only a **simplified analysis**.

For accurate real-world geographic distance, the **Haversine formula** or another geodesic distance should generally be used because the Earth is approximately spherical.

However, for a small geographic region such as the points in this dataset, Euclidean distance can be used as an approximation when the purpose is simply to demonstrate clustering concepts.

11. Conclusion

Euclidean and Manhattan distances measure similarity differently.

Euclidean distance calculates the straight-line distance between two points, while **Manhattan distance** calculates the sum of the absolute coordinate differences.

For the given geographic dataset, changing the distance metric can change the calculated similarity between points and therefore influence **cluster formation and cluster assignments**.

Euclidean distance is generally suitable for direct geometric proximity and is commonly used in K-Means. Manhattan distance can be useful when coordinate-wise movement or robustness to large differences is important.

Therefore, the selection of an appropriate distance metric is an important step in **data mining and clustering**, because it directly influences the quality and interpretation of the resulting geographic clusters.

Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

1. Introduction

Hierarchical clustering is an **unsupervised data mining technique** used to organize data into a hierarchy of clusters. It is mainly divided into two approaches:

1. **Agglomerative Hierarchical Clustering**
2. **Divisive Hierarchical Clustering**

The given dataset contains retail products characterized by their **Sales** and **Purchase Frequency**. The objective is to compare both approaches and determine their suitability for this dataset.

2. Dataset

Product	Sales (₹)	Frequency
P1	50,000	20
P2	30,000	10
P3	52,000	22
P4	25,000	8

P5	32,000	12
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The two attributes are on different numerical scales. Therefore, **normalization** should preferably be applied before distance-based clustering so that Sales does not dominate Frequency.

3. Agglomerative Clustering

Agglomerative clustering follows a **bottom-up approach**.

Initially, every product is considered as an individual cluster. The algorithm then repeatedly merges the most similar clusters until all products form one cluster.

Steps

1. Start with each product as a separate cluster.
2. Calculate the distance between products.
3. Find the two most similar clusters.
4. Merge them.
5. Recalculate the distances.
6. Continue until all products belong to one cluster.
7. Use a **dendrogram** to determine the required number of clusters.

Application to the Dataset

The products show the following patterns:

- **P1 = (50,000, 20)**
- **P3 = (52,000, 22)**

These have very similar sales and frequency values, so they are likely to form a cluster.

Similarly:

- **P2 = (30,000, 10)**
- **P5 = (32,000, 12)**

are very similar and are likely to form another cluster.

P4 has the lowest sales and frequency and may form a separate group or join the P2/P5 cluster depending on the distance and linkage method.

A possible segmentation is:

Cluster	Products	Interpretation
Cluster 1	P1, P3	High-sales, high-frequency products
Cluster 2	P2, P5	Medium-sales, medium-frequency products
Cluster 3	P4	Low-sales, low-frequency product

The exact grouping depends on the selected distance measure and linkage method.

4. Divisive Clustering

Divisive clustering follows a **top-down approach**.

Initially, all products are placed in a single cluster. The algorithm then divides the cluster into smaller groups until the desired hierarchy is obtained.

Steps

1. Start with all products in one cluster.
2. Identify the most dissimilar products or groups.
3. Split the cluster into smaller clusters.
4. Continue splitting the resulting clusters.
5. Stop when the required number of clusters is obtained.

For the given dataset, the first division may separate the **high-sales products (P1, P3)** from the lower-sales products. Further divisions can separate P4 from P2 and P5 if required.

5. Agglomerative vs Divisive Clustering

Feature	Agglomerative	Divisive
Approach	Bottom-up	Top-down
Initial state	Each product is a separate cluster	All products form one cluster
Main operation	Merging	Splitting
Dendrogram	Can represent merging hierarchy	Can represent splitting hierarchy
Computation	Generally less expensive	Generally more expensive
Large datasets	More practical than divisive	Less practical
Implementation	More commonly used	Less commonly used
Cluster formation	Similar clusters are progressively merged	Dissimilar groups are progressively separated
Reversal of decisions	Early merges are generally not reversed	Early splits are generally not reversed
Suitability for this dataset	Highly suitable	Suitable but less efficient

6. Performance Comparison

For this dataset, there are only **five products**, so both approaches can be performed easily.

Agglomerative Performance

Agglomerative clustering starts with five individual clusters and progressively merges them. Since the dataset is small, the distance calculations are simple and the hierarchy can be easily interpreted.

Divisive Performance

Divisive clustering starts with one cluster containing all five products and repeatedly divides it. This approach can require more analysis to determine the best split at each stage.

Therefore, for this small dataset, both methods are feasible, but **agglomerative clustering is generally simpler to implement and interpret.**

7. Suitability for the Given Dataset

Agglomerative clustering is more suitable for this dataset because:

1. Small Dataset

There are only five products, so the bottom-up merging process is simple.

2. Clear Similarities

P1 and P3 have very similar high sales and frequencies.

P2 and P5 also have similar sales and frequencies.

This makes the merging process intuitive.

3. Easy Visualization

A dendrogram can clearly show which products are similar and at what distance they are merged.

4. Flexible Cluster Selection

The organization can cut the dendrogram at different levels to obtain two, three, or more product segments.

5. Business Interpretation

The resulting clusters can represent high-performing, medium-performing, and low-performing products.

8. Business Interpretation of the Clusters

The clusters can help a retailer make better decisions.

High-Sales/High-Frequency Cluster

Products such as **P1 and P3** can be considered high-performing products.

The retailer can:

- Maintain sufficient inventory.
- Promote complementary products.
- Increase availability.

- Include them in premium campaigns.

Medium-Sales/Medium-Frequency Cluster

Products such as **P2 and P5** may represent moderately performing products.

The retailer can:

- Use targeted promotions.
- Offer discounts or bundles.
- Study customer preferences.
- Attempt to increase purchase frequency.

Low-Sales/Low-Frequency Cluster

P4 has the lowest sales and frequency.

The retailer can:

- Investigate why demand is low.
- Consider promotional campaigns.
- Review pricing.
- Decide whether inventory should be reduced.

9. Advantages and Limitations

Agglomerative Advantages

- Easy to understand.
- Provides a hierarchical structure.
- Does not require the final number of clusters initially.
- Dendrogram provides useful visualization.
- Suitable for small and medium-sized datasets.

Agglomerative Limitations

- Computationally expensive for very large datasets.
- Early merging decisions generally cannot be undone.
- Sensitive to the selected distance and linkage method.

Divisive Advantages

- Starts with the complete dataset and progressively identifies major divisions.
- Can reveal high-level groups first.
- Useful when the major separation between groups is important.

Divisive Limitations

- Generally more computationally expensive.
- More complex to implement.

- Less commonly used for large datasets.

10. Conclusion

Both **agglomerative and divisive clustering** are hierarchical clustering techniques, but they work in opposite directions.

Agglomerative clustering follows a **bottom-up approach**, beginning with individual products and merging similar products. **Divisive clustering** follows a **top-down approach**, beginning with all products in one cluster and repeatedly splitting them.

For the given retail dataset, **agglomerative clustering is more suitable** because the dataset is small and contains clearly similar products such as **P1–P3** and **P2–P5**. It also provides an easily interpretable dendrogram for selecting the required number of product segments.

The resulting clusters can help the retailer identify **high-performing, medium-performing, and low-performing products**, supporting inventory management, marketing, pricing, and sales decisions.

Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

1. Introduction

Clustering is an important **unsupervised data mining technique** used to group similar objects without predefined class labels.

The given dataset contains **mixed attributes**:

- **Price (₹)** → Numerical
- **Category** → Categorical
- **Rating** → Numerical

The objective is to compare **centroid-based clustering** and **density-based clustering** and analyze their effectiveness for this type of data.

2. Dataset

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5

I5	520	A	4.6
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The dataset contains two numerical attributes and one categorical attribute. Therefore, directly applying a standard numerical distance measure such as Euclidean distance to all three attributes is inappropriate unless the categorical attribute is properly encoded or a mixed-data distance measure is used.

3. Centroid-Based Clustering

Centroid-based clustering groups data points around representative central points called **centroids**.

The most common centroid-based algorithm is **K-Means**.

Working of K-Means

1. Select the number of clusters, K.
2. Initialize K centroids.
3. Calculate the distance between each data point and each centroid.
4. Assign each point to the nearest centroid.
5. Recalculate the centroids.
6. Repeat until the clusters stabilize.

Application to the Dataset

Considering the numerical attributes, the data shows two obvious patterns:

- **Category A:** I1, I3, I5 have high prices and high ratings.
- **Category B:** I2, I4 have lower prices and lower ratings.

Thus, a possible segmentation is:

Cluster Items General Characteristics

Cluster 1 I1, I3, I5 High price, high rating, Category A

Cluster 2 I2, I4 Lower price, lower rating, Category B

However, standard K-Means cannot directly process the categorical **Category** attribute. It must first be converted into numerical form, such as **one-hot encoding**, or handled using a suitable mixed-data clustering method.

4. Density-Based Clustering

Density-based clustering identifies groups based on areas where data points are densely concentrated.

A popular algorithm is **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**.

DBSCAN uses two important parameters:

- **ϵ (Epsilon):** Neighborhood radius.
- **MinPts:** Minimum number of points required to form a dense region.

Working of DBSCAN

1. Select ϵ and MinPts.
2. Identify neighboring points within ϵ .
3. Mark points in sufficiently dense regions as core points.
4. Expand clusters from connected core points.
5. Identify isolated points as noise or outliers.

Unlike K-Means, DBSCAN does not require the number of clusters to be specified in advance.

5. Handling Mixed Attributes

The main challenge in this dataset is the presence of both **numerical and categorical attributes**.

Centroid-Based Clustering

Standard K-Means works directly with numerical data.

For the categorical attribute:

Category A and B cannot simply be treated as ordinary numbers, because assigning A=1 and B=2 would incorrectly imply that B is "greater" than A.

Therefore, Category must be appropriately encoded or a clustering algorithm designed for mixed data should be used.

Density-Based Clustering

Standard DBSCAN also relies on a distance measure, so it cannot automatically solve the mixed-data problem.

A suitable distance function, such as **Gower distance**, can be used because it can handle numerical and categorical attributes together.

Therefore, both approaches require appropriate preprocessing or distance measures when mixed data is present.

6. Comparison

Feature	Centroid-Based	Density-Based
Example algorithm	K-Means	DBSCAN
Basic principle	Distance from centroid	Density of neighboring points
Number of clusters	Usually specified in advance	Not required in advance
Cluster shape	Best for compact/spherical clusters	Can detect irregular shapes
Outlier handling	Sensitive to outliers	Can identify noise/outliers
Numerical data	Very suitable	Suitable
Categorical data	Requires encoding/modification	Requires suitable distance measure
Mixed data	Needs preprocessing	Needs mixed-data distance
Scalability	Generally high	Can be good, depending on implementation
Main parameters	K	ϵ and MinPts
Interpretation	Simple	More complex

7. Effectiveness on the Given Dataset

The dataset contains only **five items**, so both approaches can be applied easily after suitable preprocessing.

Centroid-Based Effectiveness

K-Means would work well if the numerical attributes are normalized and the categorical attribute is appropriately represented.

The data shows a fairly clear separation:

- I1, I3, I5 → high price and high rating
- I2, I4 → lower price and lower rating

Therefore, centroid-based clustering can efficiently identify these groups.

Density-Based Effectiveness

DBSCAN could also identify the dense groups, especially if I1, I3 and I5 form a dense region and I2 and I4 form another.

However, with only five data points, density-based clustering can be sensitive to the choice of **ϵ and MinPts**. A poor parameter choice may classify points as noise or fail to form meaningful clusters.

Therefore, for this particular small and relatively well-separated dataset, **centroid-based clustering is likely to be more straightforward.**

8. Example of Customer/Product Interpretation

The clusters can provide useful business information.

Cluster 1 – Premium Products

I1, I3 and I5 belong to Category A and have:

- Higher prices
- High ratings

These can be considered **premium/high-quality products.**

The organization can:

- Promote them as premium products.
- Maintain sufficient inventory.
- Recommend them to customers looking for high-rated products.
- Use them in premium marketing campaigns.

Cluster 2 – Budget Products

I2 and I4 have:

- Lower prices
- Lower ratings
- Category B

These can represent **budget-oriented products.**

The organization can:

- Offer discounts.
- Improve product quality.
- Target price-sensitive customers.
- Analyze customer feedback.

9. Advantages and Limitations

Centroid-Based Clustering

Advantages:

- Simple and easy to understand.
- Computationally efficient.
- Suitable for clearly separated numerical clusters.
- Scales well for large datasets.

Limitations:

- Requires the number of clusters.
- Sensitive to outliers.
- Standard K-Means cannot directly handle categorical attributes.
- Assumes relatively compact clusters.

Density-Based Clustering

Advantages:

- Does not require the number of clusters beforehand.
- Can identify irregularly shaped clusters.
- Can detect noise and outliers.
- Useful when clusters have different shapes.

Limitations:

- Sensitive to ϵ and MinPts.
- Standard DBSCAN requires an appropriate distance measure.
- Can struggle when cluster densities vary significantly.
- Mixed attributes require a suitable distance function.

10. Conclusion

Centroid-based and density-based clustering use different approaches to identify groups.

Centroid-based clustering, such as K-Means, groups objects according to their distance from cluster centroids and is efficient for well-separated numerical data. **Density-based clustering**, such as DBSCAN, identifies dense regions and can detect irregular clusters and outliers.

For the given mixed dataset, neither standard K-Means nor standard DBSCAN should directly use the categorical **Category** attribute as an ordinary numerical value. Appropriate preprocessing, encoding, or a mixed-data distance measure such as **Gower distance** should be used.

For this small dataset, the data shows relatively clear groups, with **I1, I3, I5** forming a high-price/high-rating group and **I2, I4** forming a lower-price/lower-rating group. Therefore, **centroid-based clustering is more straightforward for this particular dataset**, while density-based clustering becomes more attractive when the data contains irregular cluster shapes or significant outliers.

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

